MASLD snRNA-seq re-analysis — methods & concepts reference

COMPUTATIONAL GENOMICS 76553 · HUJI · ONE LINE PER TERM

COUNTS, NORMALIZATION & DGE

Pseudobulk — sum all of one donor's hepatocyte UMIs into a single profile, so the donor (not the cell) is the statistical unit.

CPM — each gene as a share of the donor's total counts; simple but *compositional* — if a few genes surge, everyone else's share shrinks.

TMM — pick a reference sample, take each library's gene-wise log-ratios to it, trim the extreme movers, average the rest → the scaling factor, so surging genes can't distort it (composition-robust).

edgeR — the R/Bioconductor package that runs our per-gene differential-expression test on the pseudobulk counts.

NB quasi-likelihood GLM — edgeR's engine: model counts as negative-binomial (mean + overdispersion), fit a quasi-likelihood GLM that estimates the extra variability robustly → calibrated p-values with few replicates.

filterByExpr — keep a gene only if it has enough reads in enough samples of the smallest group (≈21k kept); low-count genes have no power and only inflate multiple testing.

Binomial down-sampling — randomly thin every nucleus to a common 1,500 UMIs so sequencing depth can't drive detection.

GENE-SET TESTS & STATISTICS

GSEA (pre-rank) — rank every gene by its fibrosis-trend statistic, then test whether a set clusters near one end instead of being scattered.

NES (GSEA) — GSEA's effect size: how strongly a set piles up at the up/down end of the ranking vs random; more negative = a stronger coordinated down-shift of the whole set.

CAMERA — a *competitive* set test: is this set shifted more than the rest of the genome? (corrects for gene–gene correlation).

ROAST / mroast — a *self-contained* set test: does the set move away from zero on its own (rotation p-value)? — ignores the rest of the genome, more conservative.

competitive vs self-contained — "more shifted than other genes?" vs "different from no change at all?" — a set can pass one and fail the other.

TOST — an *equivalence* test: not "is there a difference?" but "is it small enough to rule out anything bigger than a bound?" — lets us say "no large change."

Cramér's V — a 0–1 score of association between two categorical variables (e.g. run vs source): 0 = independent, 1 = perfectly confounded.

Spearman ρ — correlation of *ranks* (any monotonic trend, robust to outliers), from −1 to +1; ρ = −0.48 = a moderate downward trend (detox falls as fibrosis rises).

p-value — the chance of a trend this strong if nothing were really there (pure noise); p = 0.003 → ~3 in 1,000, very unlikely a fluke. Usual bar: $p < 0.05$.

FDR (q-value) — a p-value corrected for testing thousands of genes at once; FDR = 0.05 → ~5% of the called "hits" are expected to be false. Why a gene must clear FDR, not raw p.

Partial correlation — correlation between two variables after removing a third's effect (e.g. detox vs fibrosis, controlling for age).

TOOLS & DATA TERMS

decontX — models each droplet as native RNA + a shared ambient "soup," estimates each cell's ambient fraction, and subtracts it → a decontaminated matrix.

SCT — why we dropped it — Seurat normalization that swaps raw UMIs for smoothed values. Our questions are defined on integer molecules: a "≥2 UMI" call is meaningless on residuals, and ambient detection must separate 0/1/2 true molecules — SCT pools/imputes across cells and would manufacture co-expression. So we use raw counts.

Ambient RNA ("soup") — cell-free RNA from lysed cells in the droplet suspension; every droplet absorbs some, so genes from abundant cell types leak into others.

Ductular reaction — expansion of bile-duct/cholangiocyte cells in cirrhosis — it makes the soup richer in biliary RNA exactly at F4.

Doublets — two nuclei in one droplet read as a single "cell" — a hepatocyte+cholangiocyte doublet fakes co-expression.

BCV / dispersion — biological coefficient of variation: how much a gene varies beyond Poisson noise; edgeR estimates it to avoid false positives.

Anchor classification — a yes/no detection call: a nucleus is "PC" if it detects GLUL/CYP3A4 at ≥2 UMI — robust to a level drop, which is why the fraction can stay flat while the program dims.

DISTINCTIONS WE KEPT CONFUSING

Dimming vs de-zonation — de-zonation = cells change class (no signal in matched biopsies); dimming = cells keep their class, the pericentral program's level drops. Radio: genre change vs volume down.

Class gates vs the program — classification is a detection call on the gating anchors GLUL+CYP3A4 (individually flat); the program dims more broadly (detox genes CYP2E1, CYP1A2… and the pericentral landmark set), so the class stays put while the program's level falls.

Depletion vs turn-off — depletion = an anchor (pole) box shrinks; turn-off = the null (double-negative) box grows — different boxes, opposite directions.

Compression vs dimming — compression = the gradient's shape flattens (cells drift to the middle); dimming = the level drops with the shape kept.

Relative vs absolute — a z-score/correlation reads the gradient's shape (scale-free); only counting molecules reads how many cells sit where.

Donor as replicate — the ~47 donors are the independent units; cells within a donor are pseudoreplicates → summarize per donor (SE ≈ SD/√n).

Equivalence vs proof — TOST excludes shifts > ±19 pp; a ≤10-pp drift stays possible — "no large change," not "exactly zero."

Rigor & honest reading — the audit, the figures & the claims

WHAT WE CHECKED · HOW WE DISPLAY IT · WHAT WE CAN (AND CAN'T) SAY

THE CONFOUNDER BATTERY, DECODED — WHAT IT IS · WHAT WE DID / THE NUMBERS · THE VERDICT

Tissue source — stage is tangled with how tissue was obtained (healthy/end-stage = deceased-donor cubes & explants; F0–F4 = needle biopsies); the ends aren't acquisition-matched → **analyze biopsy-only F1–F4.**

Procurement stress — handling switches on stress genes that could mimic disease; immediate-early + heat-shock spike ~18× in hepatocytes *and* in endothelium (no zonation program) → **organ-wide handling, not zonation.**

Sequencing batch — runs could line up with stage; run×source Cramér's V = 0.84 (high) but run×fibrosis within biopsies = 0.40 → **ends confounded, biopsy axis estimable.**

Lobe (R/C/L) — explants span several lobes; marker detection matches across right/caudate/left → **lobe-invariant.**

Sequencing depth — deeper nuclei detect more, faking "more zonation"; down-thin to 1,000/1,500/3,000 UMIs — PC-share flat at every budget (SD 0.006–0.010) → **not depth.**

Ambient RNA — abundant-gene soup (e.g. ALB) could fake co-expression; corr(ALB share, dual fraction) over 38 donors = +0.04 → **does not drive co-expression.**

Cholangiocyte mis-annotation — mislabeled cholangiocytes would fake biliary signal; $KRT19^+ \le 0.1\%$, $EPCAM^+ \le 0.3\%$ in every anchor class → **not contaminating cholangiocytes.**

Ploidy / complexity — cell complexity could track stage and bias detection; mean nFeatures by stage 2218/3063/3169/2641/1984 is non-monotone → **does not track stage.**

Depth-match discard — thinning drops cells below budget; if PC-biased it would skew — the 3.5–7.8% dropped are uniformly lower-detection → **depth-driven, not PC-biased.**

Clinical covariates — demographics could explain trends; partial corr(detox, fibrosis | age) = −0.32 (age–fibrosis = +0.43) → **demographics don't explain it.**

WHAT THE FIGURE MUST SHOW

Dots = every donor — the spread and the outliers, shown. Nothing hidden behind a summary.

Bar = precision of the center — how well the *mean* is pinned — **not** the spread again (the dots already show that).

THE ERROR-BAR MENU

SD / IQR / whiskers — spread of donors (descriptive). Redundant with the dots — and median + IQR is literally the box plot's box.

SEM = SD/√n — precision of the mean; **tight** because of the ÷√n. ← **what we use:** a big donor spread still gives a well-pinned mean.

95% CI = SEM × t — also precision, but t ≈ 3.2 at n = 4 blows it up → reads "uncertain" and fights the message.

MEAN VS MEDIAN

Use the mean — the inference (TOST, SE ≈ 4 pp) is mean-based; a median center would describe a different quantity than the test.

Not median + IQR — robust, but it's the descriptive box again: it shows spread (the dots already do), not confidence.

WHY IT'S HONEST, NOT STAT-PICKING

Raw data shown — every donor is a dot; the outlier (PP:PC ≈ 4.0) is visible, not smoothed away.

Equivalence, both sides — TOST states what *is* excluded AND what isn't — a falsifiable claim, not "we looked and found nothing."

A real change was found — the same method/data detected detox dimming ($\rho = -0.48$) → the de-zonation null is genuine, not a power failure.

Owned soft spot — F4 has 4 donors; we lead with the F1→F4 trend, not the endpoint.

THE CLAIM LADDER — WHAT WE CAN AND CAN'T SAY

✗ **"zonation is intact / preserved"** — can't prove a negative; underpowered (F4 n = 4). **Never say this.**

✓ **"no detectable de-zonation signal"** — in matched biopsies, across every count route.

✓ **"a large shift is excluded"** — a > ±19 pp F4–F1 change is rejected by equivalence testing (TOST).

✓ **"a subtle drift is not excluded"** — ≤ 10 pp, limited by F4 n = 4. Stated, not hidden.